A Primitive-Structured Digital Twin for Industrial Human–Robot Co-Manipulation

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Human and robot capabilities complement each other

Why retain the human

- adapt to new object shapes and changing instructions;
- use dexterous hands and practical judgment;
- respond to unexpected contact or task changes.

Why add the robot

- provide strength, precision, and repeatability;
- reduce fatigue and physical load;
- execute accurate motion and force under constraints.

Target application

Focus on factory tasks that change often: full automation requires too much setup, while human-only work is tiring, risky, or inconsistent.

The human is an adapting partner, not an external disturbance.

A digital twin is a live model of the physical task

Continuously updated from sensors

The model tracks the robot, human, shared object, contact, and task using synchronized motion, force/torque, robot-state, and context measurements.

Used before and during robot action

The robot tests possible actions in the model, selects a useful action, executes it through a safety layer, and compares the measured outcome with the prediction.

Working definition

A digital twin is a synchronized, predictive, and continuously corrected digital model connected to the physical human–robot task in a closed loop.

A digital twin connects five layers in one synchronized loop

1. Physical system

The human and robot jointly move a shared object in a real workspace with tools, supports, and obstacles.

2. Synchronized state

Combine robot joints/commands, force and torque, object and human motion, contact state, and task goal on one timeline.

3. Hybrid twin core

Combine object/contact physics, online parameter estimation, a human primitive/response model, and prediction uncertainty.

4. Counterfactual prediction

For each candidate robot action, predict the human response, object motion, interaction force, task progress, and risk.

5. Decision and safety

Use layers 1–4 to compare candidate actions with a task-appropriate planner/controller. Enforce physical and safety limits; update the twin from new measurements.

A task uses context to compose reusable interaction primitives

Simple definition

A **primitive** is a reusable, short part of an interaction, such as translate, rotate, hold, align, insert, or yield. The scene provides context; a sequence of primitives describes how the task is executed.

Task representation

Task execution is a context-conditioned composition of primitives.

Context includes the object, goal, obstacles, contact locations, and physical constraints.

Illustrative table-carrying decomposition

translate → rotate → align
→ translate → hold

Each primitive has a local goal, speed/force level, and human–robot role.

The same primitive can be reused across transport, insertion, and assembly.

The primitive model connects task reasoning to control

Task context geometry + goal → VLM/VLA primitive sequence → Primitive state mode + goal + role → Action-conditioned response human + object + wrench → Safe planning/control robot action

Flow: VLM/VLA interprets the task and proposes a primitive sequence; online sensing identifies the active primitive; the response model predicts consequences; the controller selects the robot action.

Initial vocabulary and model size

Start with six modes as an initial vocabulary. Evaluate on separate trials from people and task settings not used for training. Merge indistinguishable modes; add one only if it improves prediction and control on these unseen trials.

Control-facing input and output

Input: interaction history, task context, and a candidate robot action. Output: current/next primitive, future object motion, interaction wrench, and uncertainty.

A synchronized episode needs three data groups

Static context

- CAD and geometry;
- mass, inertia, material;
- grasp/contact locations;
- goal, tolerance, human role.

Time series

- robot state, command, torque;
- 6D interaction wrench;
- object pose and twist;
- human pose and contact mode.

Outcomes and labels

- success and completion time;
- human effort and force disagreement;
- safety events;
- strategy or role changes.

Initial synchronization and fusion pipeline

Hardware timestamp/clock sync → resample on a common timeline → transform to calibrated frames → fuse asynchronous measurements with an uncertainty-aware state estimator.

Sensor requirements depend on the selected factory task

Necessary measurement categories

- robot state and commanded action;
- interaction force and torque;
- shared-object motion;
- human hand/arm motion or contact state;
- workspace and safety events.

Task-derived requirements

- enough range to avoid saturation;
- enough resolution to detect meaningful motion/force changes;
- latency below the relevant interaction time scale;
- synchronized timestamps and calibrated frames.

Initial position

Define sensor count and numerical specification after selecting the benchmark. Table transport and precision insertion require different sensing; exact hardware should be chosen with the robotics team from the task requirements.

The hybrid twin core combines four complementary models

Physics and contact dynamics

Predict shared-object motion and interaction force under robot and human inputs while enforcing known geometry and mechanics.

Human primitive and response model

Estimate the current primitive and predict human/object responses conditioned on each candidate robot action.

Online system identification

Update uncertain mass, inertia, damping, contact, payload, and actuator parameters from synchronized measurements.

Learned residual and uncertainty

Capture behavior not explained by the structured models and report confidence for risk-aware planning and fallback.

Interfaces outside the core

VLM/VLA interprets the task and proposes primitive sequences. A task-appropriate planning/control method evaluates twin predictions, while an independent safety layer enforces physical constraints before execution.

Three work packages define a realistic one-year plan

WP1: Physical digital twin

Focus: sensing, synchronization, state estimation, and hybrid dynamics.
Deliverable: validated testbed and dataset.

WP2: Human response twin

Focus: primitive transitions, action-conditioned responses, and personalization.
Deliverable: prediction across people and task variations.

WP3: World model and control

Focus: use the twin dataset to learn action-conditioned rollouts for prediction, planning, and control.
Deliverable: validated rollouts and a safe physical demonstration.

Dependency between work packages

WP1 supplies synchronized physical data to WP2 and WP3. Validate the structured twin first, then learn the world model, and connect its rollouts to planning and control only after prediction and uncertainty are tested.

ASU collaboration connects the testbed, HRI, and control

My role: twin, learning, and control

Define the digital-twin architecture and sensing/model interfaces; develop system identification, response/world models, and adaptive/safe control; lead closed-loop integration.

Joint testbed and data work

Work with the students on sensor setup and calibration, synchronized logging, data collection, and experimental troubleshooting.

HRI students' complementary role

Contribute HRI testbed experience, study protocols, participant experiments, human-centered metrics, and behavior interpretation.

Joint modeling and validation

Define primitives and response variables, train and compare models, design benchmarks, and run physical closed-loop evaluations together.

Four decisions can define the first joint milestone

1. **Benchmark task:** Select one factory operation and write its goal, physical setup, task variation, and success criteria.
2. **Starting assets:** Inventory reusable robots, sensors, simulation models, datasets, and controllers at ASU and Michigan.
3. **ASU ownership:** ASU leads the testbed, human studies, data collection, system integration, and primary experimental validation.
4. **First paper milestone:** Choose a measurable initial contribution—testbed/data, primitive human model, or world-model-based control—and its target venue.

Supporting role for Michigan

Michigan can support method development, simulation studies, theoretical analysis, and independent evaluation on the shared data. A biweekly technical call and common repository keep this support aligned with ASU's experimental milestones.

Technical Background from the Previous Discussion

Human intention representations, response prediction, and initial studies

Human Intention Modeling for Physical Human–Robot Co-Manipulation

Control-relevant representations, inference methods, and an initial early stage study

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Discussion roadmap

1. **Problem definition**
   What must a control-relevant human model predict in physical co-manipulation?
2. **Lessons from the shared papers**
   What Diffusion Co-Policy, bounded-rational games, and PACE/N-PACE already establish—and what remains open.
3. **Proposed framework**
   A reusable intention representation plus robot-action-conditioned human response prediction.
4. **Path to a testable project**
   Candidate model realizations, data collection, evaluation, and a focused first study.

Reference task: Collaborative table carrying

Task in Diffusion Co-Policy

- A human and a robot act at opposite ends of a rigid table.
- They move the table to a goal while avoiding obstacles.
- Each partner applies a planar force; success requires coordinated translation and rotation.

Information available online

- table pose and velocity;
- initial pose, goal, and obstacle locations;
- recent human force commands and robot actions;
- for physical co-manipulation: add measured 6D interaction wrench and human motion.

Why an explicit human model is needed

Diffusion Co-Policy maps history to joint actions, leaving intention implicit. An explicit model exposes intent, tests candidate robot actions, and connects prediction to planning and safety. A hybrid preserves learned flexibility.

Human intention is hidden but control-relevant

What the robot observes

- object pose and velocity x_t^O;
- interaction wrench $\mathbf{w}_t^{\mathrm{int}}$ (3D force + 3D torque);
- human/robot motion and recent actions;
- task geometry, goal, and constraints c.

What remains hidden

- desired object motion or local goal;
- leader/follower role and compliance;
- effort, force, and safety preferences;
- responses to possible robot actions.

Central question

How should human intention be represented and inferred so that it improves real-time robot planning and control in physically coupled tasks?

A useful model must resolve four uncertainties

- **Behavioral multimodality**: the same state may admit several valid goals, paths, or roles.
- **Partial observability**: motion and force provide indirect, sometimes conflicting evidence.
- **Closed-loop interaction**: the human responds to the robot and may change strategy.
- **Bounded rationality**: actions may be delayed, noisy, effort-limited, or non-optimal.

Terminology

"Multimodal sensing" means multiple input types (motion, wrench, vision).
"Multimodal behavior" means multiple possible future human strategies.

Shared papers show human models benefit interaction

Diffusion Co-Policy

- Generates several possible joint force/action sequences
- Demonstrates adaptation and role switching
- Provides the physical test setting for collaborative transport

Bounded-Rational Model

- Shows joint human actions can be represented by a game model
- Bounded rationality fits observed behavior better
- Leadership and adaptation affect performance

PACE / N-PACE

- Estimates intent/cost online while treating the peer as an adapting learner
- Shows that peer modeling can improve interaction policies
- Provides a game-theoretic interface for online control

Combined insight

Human intention is modelable and useful for control; the unresolved question is which representation is sufficiently expressive, identifiable, and transferable.

A response-conditioned hierarchy for human intention

Core contribution 1: Reusable interaction primitive

Output: $z_H = (m, g, v, \rho)$, describing the current phase, local goal, speed/force level, and interaction role.

$\Downarrow$

Core contribution 2: Response to candidate robot actions

Predict $p_\psi(u_H \mid h_t, c, \tilde{u}_R, z_H)$: how the human may respond to each candidate robot action $\tilde{u}_R$; ψ denotes learned parameters.

$\Downarrow$

Two control-facing realizations

A. Cost/preferences

Response-conditioned cost $J_H(u_R)$ and preference weights w_H.

B. Desired object motion

Response-conditioned goal, path mode, or trajectory ξ_H^*.

How the intention representations will be evaluated

1. Intention identifiability

Can the model distinguish route, role, and task phase early? Measure classification accuracy, posterior confidence, and detection delay.

2. Future prediction

Measure trajectory displacement, wrench prediction error, likelihood of observed behavior, and coverage of different valid futures.

3. Closed-loop control value

Measure task success, completion time, human effort, conflicting interaction force, safety violations, and online computation time.

4. Transfer and adaptation

Test unseen users, objects, obstacle layouts, and tasks; report zero-shot performance and data required for personalization.

High-level option: Reusable interaction primitives

Switching latent-state model

$$z_t^H = (m_t, g_t, v_t, \rho_t), \qquad m_t \in \mathcal{M}_{\text{int}}$$

where $\mathcal{M}_{\text{int}}$ includes translate, rotate, hold, align, insert, and yield.

- m_t: current interaction primitive;
- g_t: local direction, pose, or contact goal;
- v_t: desired speed or interaction-force level;
- ρ_t: leader/follower role or compliance.

Why it supports generalization

Factory tasks reuse a small vocabulary: transport may be *translate–rotate–align*; insertion may be *approach–align–hold–insert*.

Initial primitive vocabulary

Start with six object-level modes: translate, rotate, hold, align, insert, and yield. Define them by motion/force patterns, not task names.

How primitives are identified and learned

Identify the current primitive

- Use task-phase labels when the phase is known.
- Otherwise use Expectation–Maximization (EM) to infer latent switching modes from object motion, wrench, and context.
- Infer $p(m_t \mid h_t, c)$ online rather than committing to one mode too early.

Decide how many primitives are needed

Begin with the six-mode vocabulary. Merge modes with indistinguishable motion/force responses; add a mode only when repeated residual patterns improve held-out prediction and control.

Learn continuous parameters

Within each mode, train goal, speed/force, and role parameters from future object motion and interaction-wrench prediction.

Shared response layer: Predict human accommodation

Passive prediction

$$p(u^H_{t:t+H} \mid h_t, c)$$

What might the human do next?

$\Longrightarrow$

Response to a candidate robot action

$$p_\psi(u^H_{t:t+H} \mid h_t, c, \tilde{u}^R_{t:t+H}, z_H)$$

How might the human respond if the robot executes $\tilde{u}^R$?

- h_t: recent interaction history; c: task context; z_H: current primitive.
- $\tilde{u}^R$: candidate robot action sequence; u^H: predicted human response.
- The output conditions Options A and B; it is not a separate high-level intention by itself.

Option A: Infer the human's objective

Task-conditioned human cost

$$J_H = \sum_{k=t}^{t+H} {w_H}^\top \phi_\psi(x_k^O, u_k^H, u_k^R, c)$$

Cost and feature notation

J_H: cumulative human cost; ϕ_ψ: learned and physical cost-feature vector; w_H: person-specific human preference weights.

State and action notation

x_k^O: object state; u_k^H: human action; u_k^R: candidate robot action considered by the planner. The sum covers steps $k = t$ through $t + H$ under context c.

Option A: Learning the cost model

Why it is attractive

The weights expose tradeoffs among progress, effort, interaction force, safety, and coordination. Because u^R enters the features, the inferred human optimum can change with the candidate robot action. The model connects directly to IOC, PACE, and N-PACE.

Training without cost labels

Hierarchical MaxEnt IOC learns shared cost features ϕ_ψ and latent person-specific weights w_H from demonstrations.

Main challenges

Different costs can explain the same action; learned features may lose physical meaning; and human behavior is not perfectly optimal.

Practical constraint

Anchor learned features to known physical quantities so that each component of w_H retains a stable interpretation online.

Option A can supply N-PACE's human model

Offline learned representation

- Train a shared task-conditioned feature map ϕ_ψ from multi-person demonstrations.
- Learn a context-dependent prior $p(w_H \mid c)$.
- Freeze ϕ_ψ at deployment so the online parameter meaning remains stable.

Online N-PACE layer

- Define $J_H = \sum_k w_H^\top \phi_\psi(x_k^O, u_k^H, u_k^R, c)$.
- N-PACE estimates w_H and how the human is updating beliefs about the robot.
- The inferred cost enters the dynamic-game solver to update robot actions.

What the learned model contributes

It replaces task-by-task manual feature design with a reusable cost representation; N-PACE retains peer-aware online inference and interaction-aware control.

Option B: Desired object motion as intention

Desired trajectory model

$$p_\psi(\xi_H^* \mid h_t, c, \tilde{u}^R_{t:t+H}), \qquad \xi_H^* = \{x_{O,k}^{H,*}\}_{k=t}^{t+H}$$

ξ_H^* is the human-intended object trajectory; $x_{O,k}^{H,*}$ is its desired object state at step k; $\tilde{u}^R$ is the candidate robot action sequence. The model outputs a distribution because several responses may be valid.

Possible outputs

- terminal pose or waypoint;
- discrete path/strategy mode;
- distribution over full desired trajectories.

Why it is attractive

Directly useful as a reference for shared-object MPC; easier to label and visualize than latent cost parameters.

Main challenges

Multiple valid trajectories; changing intention; and limited training coverage of possible robot actions.

Option B: Training and control interface

Training without a unique intent label

Use future object motion paired with the executed robot action as a self-supervised target, but learn several possible human responses rather than one "correct" path. Controlled experiments can also provide route or goal labels.

Learning objective

Use a diffusion loss or a loss that keeps the best among several predictions, so left/right routes and alternative rotation strategies are retained instead of averaged together.

Role in the complete system

The desired motion is a low-level reference for shared-object model predictive control (MPC). Combine it with a primitive or preference model when the robot must explain why the human changes strategy.

Related: Townsend et al. (2017); Franceschi et al. (2023).

Each representation serves a different modeling layer

Representation	Control interface	Key strength	Main weakness	Best use
Interaction primitive	Latent mode z_H	Cross-task structure	Needs mode discovery	Task-phase inference
Response layer	$p(u_H \mid \tilde{u}_R, z_H)$	Models accommodation	Needs diverse robot actions	Candidate-action evaluation
Cost / preferences	J_H, w_H, ϕ	Interpretable and game-compatible	Identifiability and feature design	Negotiation and tradeoffs
Desired trajectory	Reference ξ_H^*	Direct and visual	Needs robot-action coverage	Shared-object MPC

Proposed position

Condition both intention interfaces on candidate robot actions. The discrete primitive vocabulary is a tractable starting point, not a fixed ontology; future work can learn a continuous, task-conditioned latent representation across tasks without predefining the modes.

A simpler hierarchy connects intention modeling to control

Observations motion + interaction wrench + context → Interaction primitive $z_H = (m, g, v, \rho)$ → Human response model $p(u_H \mid h, c, \tilde{u}_R, z_H)$ → A–B control interface $w_H^\top \phi_\psi$ / ξ_H^* → N-PACE / MPC robot action u_R

Learned offline

Primitive transitions, shared feature/trajectory representations, and the population response prior.

Adapted online

Current primitive, desired motion, preference weights, and person-specific response uncertainty.

Data collection supplies weak and self-supervision

Minimum synchronized modalities

- object pose, velocity, and acceleration;
- 6D interaction wrench;
- robot state and commanded action;
- task goal, geometry, contact configuration;
- optional human body/arm pose and vision.

Where training signals come from

- **weak labels**: assigned goal, path condition, task phase, role instruction;
- **self-supervision**: future object motion and interaction wrench;
- **preference evidence**: choices under controlled effort/force/path tradeoffs;
- **latent learning**: EM / IOC likelihood when no direct label exists.

Practical collection plan

Use accessible human–human demonstrations for pretraining, then collect human–robot trials with diverse robot actions for interactive-response identification. A curated multimodal benchmark can itself support a dataset/benchmark paper.

Thank you

Follow-up Research Directions

Longer-term subprojects in sensing, control, and cross-task generalization

First study: Compare intention representations

Physical test setting

Collaborative object transport with:

- two feasible paths around an obstacle;
- coordinated translation and rotation around obstacles;
- changing leader/follower roles;
- synchronized object state and 6D interaction wrench.

Model comparison

1. passive cost and trajectory baselines (A–B);
2. response-conditioned cost inference;
3. response-conditioned desired trajectory;
4. hierarchical hybrid model.

Central hypothesis

Combining structured intention variables with robot-action-conditioned response prediction will improve early intent inference and closed-loop coordination.

Separate model quality from team performance

Human-model quality

- path/role classification and detection delay;
- desired-trajectory displacement error;
- force/action likelihood and calibration;
- adaptation after strategy changes;
- leave-one-person-out generalization.

Closed-loop interaction quality

- task success and completion time;
- human and robot effort;
- unnecessary interaction force;
- path efficiency and smoothness;
- safety and comfort violations.

Critical ablations

Motion vs. motion+wrench; passive vs. robot-action-conditioned; fixed vs. learned features; population prior vs. online personalization.

Proposed framework for physical co-manipulation

1. Physical interaction evidence

Object and human motion, 6D interaction wrench, task context, and candidate robot actions.

$\Downarrow$

2. Human intention and response model

Infer a reusable human representation, then predict how the human may respond to each candidate robot action. Expose either interpretable cost/preferences (Option A) or desired object motion (Option B) to the controller.

$\Downarrow$

3. Robot planning and safe execution

Use N-PACE or shared-object MPC to select robot actions under physical constraints, then apply an independent safety filter. Initial validation uses collaborative object transport.

Notation used throughout the discussion

Indices and agents

- t: current time; k: a time inside the prediction horizon;
- H: number of future steps being predicted;
- superscripts O, H, and R: object, human, and robot.

Signals

x: state or pose; u: action or force command; $\mathbf{w}^{\text{int}}$: measured interaction wrench.

Model inputs

h_t: recent interaction history up to time t; c: task context, including goal, geometry, object properties, and constraints.

Model notation

$p(a \mid b)$: distribution of a given b; ψ: learned model parameters; $*$: desired or latent quantity; $\dot{x}$: time derivative of x. PACE means Peer-Aware Cost Estimation; N-PACE is its nonlinear extension.

Offline priors support online personalization

Offline: learn shared structure

- task/interaction encoders from multi-person data;
- trajectory modes and residual cost features;
- robot-action-conditioned response model;
- uncertainty estimates tested on unseen participants;

Online: update compact variables

- belief over intent mode $b_t(z_H)$;
- desired motion parameters ξ_H^*;
- preference weights w_H;
- optional small person-specific adapter.

Additional notation used in the subprojects

Human-model learning

J_H: human cost; w_t^H: time-varying human preference weights; ϕ_ψ: learned cost features; u^H_{future} and u^R_{future}: future human and robot action sequences.

Robot planning

u_R^*: selected robot action sequence; J_R: robot planning cost; $\mathbb{E}$: average over possible human responses; $\arg\min$: the robot action that produces the smallest expected cost.

Probability model

p_ψ: learned conditional distribution; ψ: its trained parameters. The conditioning variables are written after the vertical bar $|$.

Shared inputs

x_t: system state; h_t: recent history; c: task context. Superscripts H and R denote human and robot quantities.

Bayesian personalization from physical interaction

Update equation

$$b_{t+1}(\eta_H) \propto p(y_{t+1} \mid h_t, c, u_t^R, \eta_H)\, b_t(\eta_H)$$

Belief and latent variables

$\eta_H = (z_H, \theta_H, w_H)$ collects the human primitive, desired-motion parameters, and preference weights. $b_t(\eta_H)$ is the robot's current belief over them.

New physical evidence

$y_{t+1} = (\mathbf{w}_t^{\mathrm{int,obs}}, x_{t+1}^O)$ combines the observed wrench and next object state. u_t^R is the executed robot action; $p(\cdot)$ scores how well each human model explains the evidence.

Interpretation

$\propto$ means: multiply the prior belief by the likelihood of the new observation, then normalize. Models that better predict the human–robot interaction receive more probability.

Cold start: Safe behavior before personalization

New user, no interaction history

- Initialize from a population and physical-dynamics prior.
- Adapt only compact variables: primitive, preference weights, and desired-motion parameters.
- Increase robot assistance only as confidence in the human model improves.

Initial robot behavior

- Use low-gain impedance or compliant control.
- Bound speed and interaction force; retain a safety filter.
- Apply small information-seeking actions only when they remain safe.

If no offline human dataset exists

Do not train a large response model purely online. Start from a structured dynamics/game prior and a safe default controller, collect interaction data, and learn the population model across later trials.

Backup Option C: What is an attractor model?

Intuitive meaning

An attractor model assumes the human is pulling the shared object toward an internal target. The predicted desired velocity points toward that target and decreases as the object approaches it.

Position example

$$\dot{p}_O^{H,*} = K_p\big(p_H^*(\tilde{u}_R) - p_O\big)$$

p_O: current object position; $p_H^*(\tilde{u}_R)$: inferred human target under candidate robot action $\tilde{u}_R$; $K_p > 0$: response gain; $\dot{p}_O^{H,*}$: human-desired object velocity.

Why it is attractive

The robot estimates a target and a few gains instead of assuming that the human solves a complete optimal-control problem.

What the model can and cannot express

It is well suited to local motion toward a goal, but one attractor cannot represent several competing routes or long-term effort/safety tradeoffs.

Backup Option C: Full rigid-object pose model

Response-conditioned translation and rotation

$$\begin{bmatrix} \dot{p}_O^{H,*} \\ \omega_O^{H,*} \end{bmatrix} = f_H(p_O, R_O, c, \tilde{u}_R; \theta_H)$$

p_O and R_O are the current object position and orientation; $\dot{p}_O^{H,*}$ and $\omega_O^{H,*}$ are the human-desired linear and angular velocities.

Translation component

$\dot{p}_O^{H,*} = K_p\big(p_H^*(\tilde{u}_R) - p_O\big)$, where p_H^* is the inferred target position and K_p is the translation response gain.

Rotation component

$\omega_O^{H,*} = K_{\mathrm{rot}}\, e_R(R_O, R_H^*)$, where R_H^* is the inferred target orientation and e_R converts orientation error to a 3D rotation vector.

Backup Option C: Training and online inference

Training without parameter labels

Fit the likelihood of observed object motion under diverse robot actions using constrained system identification. The desired pose and response gains are latent variables rather than manually labeled quantities.

Switching between local models

Use Expectation–Maximization (EM) offline or a particle filter online to infer θ_H and the active mode from motion and wrench measurements.

Physical constraints

Constrain f_H so the predicted motion remains stable, feasible for the shared object, and consistent with contact and actuation limits.

Related: constraint-aware dynamical-system intent estimation with particle filtering (Shao et al., 2024).

SP1: Build the physical evidence base

Theme and focus

Physically grounded multimodal data

- Build shared-object transport experiments
- Synchronize motion, object state, and force/torque
- Create ambiguous paths, role switching, and strategy changes
- Compare motion-only, wrench-only, and multimodal inference

Challenges → solutions

- No direct intent labels → hierarchical task/strategy/action labels
- Noisy contact data → filtering and contact-state estimation
- High person variability → probabilistic latent-state models
- Costly data → human–human, simulation, and HRI data mixture

Expected paper: A multimodal benchmark and systematic study of force/torque for human intent inference.

SP2: Make intent and preference explicit

Bounded-rational human model

$$J_H = \sum_t w_t^{H\top} \phi_\psi(x_t, u_t^H, u_t^R, c)$$

Reusable features may represent progress, pose, effort, interaction force, safety, smoothness, and coordination.

- Infer intent, role, preference, and rationality online
- Represent the human's belief about robot intent
- Treat the human as another adapting learner

Challenges → solutions

- Cost identifiability → Bayesian / maximum-entropy IOC
- Ambiguous intent → informative robot actions
- Unknown adaptation → latent learning-state estimation
- Online computation → local game approximations and MPC

Expected paper: Peer-aware intent and preference inference for physical co-manipulation.

SP3: Learn responses to possible robot actions

Interactive response prediction

$$p_\psi(u^H_{\text{future}} \mid h_t, c, u^R_{\text{future}})$$

- Predict multimodal human response trajectories
- Condition on motion, wrench, context, and candidate robot actions
- Predict uncertainty and latent-state transitions
- Compare Transformer, diffusion, and latent-variable models

Challenges → solutions

- Limited action coverage → diverse robot probing
- Correlation vs. response → counterfactual conditioning
- Out-of-distribution actions → uncertainty estimation and detection
- Slow inference → distilled or few-step diffusion

Expected paper: Robot-action-conditioned diffusion models for multimodal human response prediction.

SP4: Close the loop through safe control

Human-model-aware planning

$$u_R^* = \arg\min_{u_R} \mathbb{E}_{u_H \sim p_\psi}[J_R]$$

subject to object dynamics, actuation, force, collision, and safety constraints.

- Optimize progress, team effort, and interaction force
- Replan as the human model is updated
- Use a safety filter before execution

Challenges $\rightarrow$ solutions

- Prediction error $\rightarrow$ short-horizon MPC
- Distributional complexity $\rightarrow$ scenario sampling
- Excess conservatism $\rightarrow$ risk-sensitive objectives
- Hard safety limits $\rightarrow$ control-barrier safety filter

Expected paper: Uncertainty-aware planning and safe control with interactive human response models.

SP5: Transfer human concepts across tasks and people

Task-conditioned generalization

Shared human latent concepts:

- intent and local strategy;
- leader–follower role and compliance;
- effort, safety, and interaction preferences;
- reaction delay and adaptation state.

Encode goals, object dynamics, geometry, contacts, and constraints as task context. Adapt only latent states, preference weights, or a small adapter online.

Challenges → solutions

- Different task structures → object-centric states and wrenches
- Limited physical data → simulation and human–human pretraining
- Negative transfer → structured task encoders
- Unstable personalization → Bayesian updates or small adapters

Expected paper: A task-conditioned human model for cross-task co-manipulation and online personalization.

Five subprojects answer one connected question chain

SP1 — Observe: What physical signals reveal human intent and interaction strategy?

↓

SP2 — Infer: What does the human intend, prefer, and believe about the robot?

↓

SP3 — Predict: How will the human respond to candidate robot actions?

↓

SP4 — Act: How should the robot coordinate safely under prediction uncertainty?

↓

SP5 — Transfer: What representations generalize across people and tasks?

A focused transport study can test the core hypothesis

Initial task

A human and robot transport a shared object through an environment with multiple feasible paths.

Model inputs

- Object and human motion
- Interaction force/torque
- Goal and environment geometry
- Candidate robot actions

Comparisons

1. Reactive control without a human model
2. Passive prediction from interaction history
3. Structured intent inference
4. Robot-action-conditioned prediction
5. Hybrid structured + learned model

Evaluation

- Prediction accuracy and calibration
- Success, time, and team effort
- Interaction force and disagreement
- Adaptation, safety, and comfort violations

The program connects human understanding to robot action

1. A physically grounded multimodal benchmark
2. Bounded-rational, peer-aware intent inference
3. Interactive prediction of responses to robot actions
4. Safe planning over human response distributions
5. Cross-task generalization and online personalization

Long-term vision

A versatile human model that connects physical interaction, human adaptation, and robot decision-making across co-manipulation scenarios.

Proposed starting point

Use a physical collaborative-transport task to test whether force/torque sensing and robot-action-conditioned prediction improve intent inference, coordination, and safe control.

Questions to refine the research direction

- Which physical co-manipulation task is the most useful initial test case?
- What sensing and robotic platforms are currently available?
- Should the first contribution emphasize human intent inference, interactive response prediction, or planning and control?
- Which components of RISE Lab's existing game-theoretic framework can be extended directly to physical coupling?
- What level of cross-task generalization would be most valuable for the current project?

Discussion objective

Identify a focused first study that contributes to the larger goal of a generalizable and interactive human model.